



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-022 Data Ingestion & Interfaces - Protected Review Pack

Open one matching section only after frozen genuine attempt

Artifact	C3_022_DATA_INGESTION_INTERFACES_REVIEW_PACK_PROTECTED_RC1
Status	PROTECTED - ATTEMPT REQUIRED
Teaching standard	M01 beginner-first teacher loop. Videos are visual companions; the book remains sufficient fallback.
C2B boundary	Bridge foundations may be assumed, but C3 engineering behavior is demonstrated, broken, repaired, transferred and validated.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

Course rule: preserve what arrived -> validate interpretation -> quarantine ambiguity -> publish durably -> commit state last -> prove replay safety. A green job is never sufficient evidence by itself.



How to use a review without copying

A review diagnoses your reasoning. It does not replace the attempt. Compare your evidence against the invariants below, repair only the weak competency, close the review, then solve the changed task from memory. Mastery is awarded only from the changed task + validation.

Never edit or replace attempt_001 after opening this file. Save repair and changed retry as new append-only evidence.



R01 - Review C3-01-L01

FIRST CHECK

Prove raw hash/path exists; accepted + quarantine = source records; staging path is different from landing; manifest has UTC run time.

COMMON FAILURE SIGNALS

- Writing cleaned output over the source file.
- Calling staging "raw" after types/columns were already changed.
- Using only a filename such as latest.csv with no run/window identity.
- Recording accepted count but not quarantine count.

DIAGNOSE IN THIS ORDER

1) Restate input/output grain or state boundary. 2) Compare expected vs actual. 3) Run the independent control. 4) Locate the first invariant that fails. 5) Repair only that boundary. 6) Re-run the original attempt for understanding, but preserve attempt_001 unchanged.

RECOVERY

If you already edited the only copy, stop. Re-acquire the source if possible, create a new immutable landing run, and document that the earlier run lacks trustworthy raw evidence.

WHY THE WORKING PATTERN IS SOUND

The hash identifies source bytes independently of whatever your parser later decides. Landing and staging can now disagree without destroying evidence.

CLOSE REVIEW - CHANGED RETRY

After review, change source type and one contract field. Rebuild from blank without looking at the reviewed attempt.

MASTER WHEN

Changed retry succeeds without this review open; independent control passes; you can teach the concept and one failure/recovery path without reading.



R02 - Review C3-01-L02

FIRST CHECK

accepted + quarantine = source data rows; required headers present; keys unique where required; two manual raw-to-output comparisons; quoted delimiter preserved.

COMMON FAILURE SIGNALS

- Using ``line.split(",")``.
- Assuming blank numeric value means 0.
- Dropping bad rows without raw row + reason.
- Trusting parsed row count without source accounting.

DIAGNOSE IN THIS ORDER

1) Restate input/output grain or state boundary. 2) Compare expected vs actual. 3) Run the independent control. 4) Locate the first invariant that fails. 5) Repair only that boundary. 6) Re-run the original attempt for understanding, but preserve `attempt_001` unchanged.

RECOVERY

If widths or values shift, inspect ``reader.fieldnames``, ``reader.line_num``, the raw line and dialect assumptions. Do not "fix" output values before preserving the bad source row.

WHY THE WORKING PATTERN IS SOUND

``csv.DictReader`` applies CSV quoting/newline rules and exposes fields by name, so you can validate schema and convert types deliberately.

CLOSE REVIEW - CHANGED RETRY

Use a different delimiter/blank pattern and rebuild without opening the review.

MASTER WHEN

Changed retry succeeds without this review open; independent control passes; you can teach the concept and one failure/recovery path without reading.



R03 - Review C3-01-L03

FIRST CHECK

parent output count = source orders; child count = total source items; each child has parent key; parent total reconciles once per order.

COMMON FAILURE SIGNALS

- Flattening parent + child into one table then summing parent measures.
- Losing parent key on child rows.
- Treating missing array and empty array as identical without contract.
- Not recording source/run identity.

DIAGNOSE IN THIS ORDER

1) Restate input/output grain or state boundary. 2) Compare expected vs actual. 3) Run the independent control. 4) Locate the first invariant that fails. 5) Repair only that boundary. 6) Re-run the original attempt for understanding, but preserve attempt_001 unchanged.

RECOVERY

If totals inflate, stop using the flattened output. Restate parent and child grain on paper, regenerate separate outputs, then reconcile before business aggregation.

WHY THE WORKING PATTERN IS SOUND

The control counts come directly from source structure before flattening, so they can detect accidental row multiplication.

CLOSE REVIEW - CHANGED RETRY

Normalize a fresh customer->addresses document into customer and address grains.

MASTER WHEN

Changed retry succeeds without this review open; independent control passes; you can teach the concept and one failure/recovery path without reading.



R04 - Review C3-01-L04

FIRST CHECK

PyArrow version recorded; file opens as Parquet; metadata row count = projected row count; projection contains exactly requested columns; schema difference explained.

COMMON FAILURE SIGNALS

- Renaming CSV to `.parquet``.
- Loading every column before checking schema.
- Assuming "Parquet" guarantees compatible types across files.
- Claiming runtime evidence if PyArrow never ran.

DIAGNOSE IN THIS ORDER

1) Restate input/output grain or state boundary. 2) Compare expected vs actual. 3) Run the independent control. 4) Locate the first invariant that fails. 5) Repair only that boundary. 6) Re-run the original attempt for understanding, but preserve `attempt_001` unchanged.

RECOVERY

If import fails, confirm the active venv and ``python -m pip show pyarrow``. If installation is blocked, record the blocker; do not fabricate Parquet evidence. If schema differs, inspect metadata before coercing.

WHY THE WORKING PATTERN IS SOUND

Parquet embeds schema and row-group metadata. PyArrow can inspect that structure and perform column projection without pretending the file is opaque row text.

CLOSE REVIEW - CHANGED RETRY

Build a new Parquet from a changed CSV and inspect one different two-column projection.

MASTER WHEN

Changed retry succeeds without this review open; independent control passes; you can teach the concept and one failure/recovery path without reading.



R05 - Review C3-01-L05

FIRST CHECK

timeout set; expected status checked; JSON/content checked; required keys checked; token absent from source/log; missing-auth case fails clearly.

COMMON FAILURE SIGNALS

- No timeout.
- Calling `.json()` and trusting shape blindly.
- Logging Authorization header.
- Treating 401 as retryable transient failure.

DIAGNOSE IN THIS ORDER

1) Restate input/output grain or state boundary. 2) Compare expected vs actual. 3) Run the independent control. 4) Locate the first invariant that fails. 5) Repair only that boundary. 6) Re-run the original attempt for understanding, but preserve `attempt_001` unchanged.

RECOVERY

If connection fails, verify server terminal and `/health` first. If 401, inspect presence of environment token without printing it. If JSON shape is wrong, fail before landing as accepted data.

WHY THE WORKING PATTERN IS SOUND

Timeout bounds waiting; status/media/shape checks separate transport success from data-contract success; environment config prevents token hard-coding.

CLOSE REVIEW - CHANGED RETRY

Use another mock endpoint and define its contract from scratch.

MASTER WHEN

Changed retry succeeds without this review open; independent control passes; you can teach the concept and one failure/recovery path without reading.



R06 - Review C3-01-L06

FIRST CHECK

landed page count known; final unique IDs = expected full set; no repeated page/cursor cycle; 429 recovery bounded; token secret scan clean.

COMMON FAILURE SIGNALS

- Stopping after first 200 response.
- Constructing guessed next URLs instead of following documented continuation.
- Retrying 429 immediately in a tight loop.
- Writing token into committed config.

DIAGNOSE IN THIS ORDER

1) Restate input/output grain or state boundary. 2) Compare expected vs actual. 3) Run the independent control. 4) Locate the first invariant that fails. 5) Repair only that boundary. 6) Re-run the original attempt for understanding, but preserve attempt_001 unchanged.

RECOVERY

If IDs repeat, log page/cursor metadata and inspect continuation logic. If 429 repeats, inspect Retry-After and retry bounds. If token fails, diagnose auth separately from pagination.

WHY THE WORKING PATTERN IS SOUND

Pagination termination is part of the source contract; unique-ID/full-count controls detect missed/repeated pages; throttling is handled deliberately rather than hammering the producer.

CLOSE REVIEW - CHANGED RETRY

Use a cursor-style changed mock case and re-prove complete unique extraction.

MASTER WHEN

Changed retry succeeds without this review open; independent control passes; you can teach the concept and one failure/recovery path without reading.



R07 - Review C3-01-L07

FIRST CHECK

retry class explicit; max attempts explicit; delay capped; jitter present; success path and bounded-failure path both evidenced; permanent error not retried.

COMMON FAILURE SIGNALS

- Infinite retry loop.
- Retrying 400/401/schema failures blindly.
- No jitter across many workers.
- Retrying a non-idempotent write without an idempotency strategy.

DIAGNOSE IN THIS ORDER

1) Restate input/output grain or state boundary. 2) Compare expected vs actual. 3) Run the independent control. 4) Locate the first invariant that fails. 5) Repair only that boundary. 6) Re-run the original attempt for understanding, but preserve attempt_001 unchanged.

RECOVERY

If retries never stop, add explicit attempt/time caps first. If permanent failures are being retried, split transport/transient policy from validation/auth errors.

WHY THE WORKING PATTERN IS SOUND

Classification prevents useless retry; exponential delay reduces pressure; jitter desynchronizes clients; bounded attempts guarantee the pipeline eventually returns control.

CLOSE REVIEW - CHANGED RETRY

Change retryable status and max budget, then re-run without review open.

MASTER WHEN

Changed retry succeeds without this review open; independent control passes; you can teach the concept and one failure/recovery path without reading.



R08 - Review C3-01-L08

FIRST CHECK

explicit columns; bound values; boundary stated; stable ordering; source count equals output; key uniqueness checked.

COMMON FAILURE SIGNALS

- `SELECT \*` as a long-lived contract.
- String-concatenating timestamps/IDs into SQL.
- Using only timestamp when ties can exist.
- Changing source query and control query together so both share same bug.

DIAGNOSE IN THIS ORDER

1) Restate input/output grain or state boundary. 2) Compare expected vs actual. 3) Run the independent control. 4) Locate the first invariant that fails. 5) Repair only that boundary. 6) Re-run the original attempt for understanding, but preserve attempt_001 unchanged.

RECOVERY

If counts differ, print the lower/upper boundary, run source control independently, inspect equal-to-boundary rows and confirm time-zone/type semantics before widening the window.

WHY THE WORKING PATTERN IS SOUND

Explicit boundary defines eligibility; parameter binding separates data from query text; stable ordering makes repeated extraction comparison deterministic.

CLOSE REVIEW - CHANGED RETRY

Use a new window/table and rebuild query + independent control from blank.

MASTER WHEN

Changed retry succeeds without this review open; independent control passes; you can teach the concept and one failure/recovery path without reading.



R09 - Review C3-01-L09

FIRST CHECK

every batch has unique path + manifest + hash + count + source window; changed bytes yield changed fingerprint; original remains preserved.

COMMON FAILURE SIGNALS

- One `latest.csv` path.
- Using filesystem modification time as business event time.
- No hash or count.
- Treating changed historical bytes as normal rerun without alert/evidence.

DIAGNOSE IN THIS ORDER

1) Restate input/output grain or state boundary. 2) Compare expected vs actual. 3) Run the independent control. 4) Locate the first invariant that fails. 5) Repair only that boundary. 6) Re-run the original attempt for understanding, but preserve attempt_001 unchanged.

RECOVERY

If prior batches were overwritten, reconstruct from source if available. From now on use immutable paths and record the evidence gap rather than inventing old fingerprints.

WHY THE WORKING PATTERN IS SOUND

Window identity answers what data the batch represents; run ID answers when/how you processed it; fingerprint proves whether source bytes changed.

CLOSE REVIEW - CHANGED RETRY

Land a changed-source historical batch under a new run and explain why it must not overwrite old evidence.

MASTER WHEN

Changed retry succeeds without this review open; independent control passes; you can teach the concept and one failure/recovery path without reading.



R10 - Review C3-01-L10

FIRST CHECK

old/new cursor logged; every eligible tuple $>$ old and $\leq$ new; source/output counts reconcile; state unchanged on failure; rerun stable.

COMMON FAILURE SIGNALS

- Timestamp-only cursor.
- Advancing state before output validation.
- Using current clock time as source watermark.
- No proof of what happens on replay.

DIAGNOSE IN THIS ORDER

1) Restate input/output grain or state boundary. 2) Compare expected vs actual. 3) Run the independent control. 4) Locate the first invariant that fails. 5) Repair only that boundary. 6) Re-run the original attempt for understanding, but preserve attempt_001 unchanged.

RECOVERY

If records were skipped, compare old state with source rows around the boundary. Restore last known safe committed state, replay from there, deduplicate by declared key, and document recovery.

WHY THE WORKING PATTERN IS SOUND

Tuple ordering gives a total deterministic boundary across ties. Committing state after durable validation prevents the cursor from jumping past unpublished data.

CLOSE REVIEW - CHANGED RETRY

Use a different tie key and changed two-run dataset; prove same invariants.

MASTER WHEN

Changed retry succeeds without this review open; independent control passes; you can teach the concept and one failure/recovery path without reading.

R11 - Review C3-01-L11

FIRST CHECK

positions ordered/unique as required; each op handled; final state manually predicted; replay yields identical final state; raw log unchanged.

COMMON FAILURE SIGNALS

- Treating delete like an update with null fields without contract.
- Deduplicating different updates just because key repeats.
- Ignoring source ordering.
- Installing Kafka now even though it is not needed for event-semantic mastery.

DIAGNOSE IN THIS ORDER

1) Restate input/output grain or state boundary. 2) Compare expected vs actual. 3) Run the independent control. 4) Locate the first invariant that fails. 5) Repair only that boundary. 6) Re-run the original attempt for understanding, but preserve attempt_001 unchanged.

RECOVERY

If final state looks wrong, replay one event at a time and log position/key/op/current state. Find first divergence instead of patching the final table.

WHY THE WORKING PATTERN IS SOUND

Source position defines event order; key identifies target row; op determines state transition. Keeping raw events separate lets you replay/audit.

CLOSE REVIEW - CHANGED RETRY

Use a changed event sequence including delete/recreate; rebuild from raw events without review.

MASTER WHEN

Changed retry succeeds without this review open; independent control passes; you can teach the concept and one failure/recovery path without reading.

R12 - Review C3-01-L12

FIRST CHECK

schema diff recorded; policy decision explicit; UTC conversion proved; ambiguous time retained in quarantine; critical unresolved drift prevents unsafe state advance.

COMMON FAILURE SIGNALS

- Automatically accepting any added/removed/renamed field.
- Using machine local timezone implicitly.
- Replacing bad timestamp with current time.
- Advancing state after critical drift blocked records unpredictably.

DIAGNOSE IN THIS ORDER

1) Restate input/output grain or state boundary. 2) Compare expected vs actual. 3) Run the independent control. 4) Locate the first invariant that fails. 5) Repair only that boundary. 6) Re-run the original attempt for understanding, but preserve attempt_001 unchanged.

RECOVERY

Stop publication/state advance; preserve raw batch; record observed schema/time ambiguity; get producer/contract clarification; update versioned policy; replay from unchanged checkpoint.

WHY THE WORKING PATTERN IS SOUND

Schema policy separates structural change from semantic guesswork; aware timestamps map to an instant; quarantine preserves ambiguous records without inventing meaning.

CLOSE REVIEW - CHANGED RETRY

Use a changed schema/time case; classify it without copying prior decisions.

MASTER WHEN

Changed retry succeeds without this review open; independent control passes; you can teach the concept and one failure/recovery path without reading.



R13 - Review C3-01-L13

FIRST CHECK

state_before captured; state unchanged on injected failure; rerun accepted keys stable; temp state not left as active checkpoint; final state equals last durable accepted boundary.

COMMON FAILURE SIGNALS

- Checkpoint first, output second.
- Appending duplicate outputs on every replay.
- Catching exception then returning success.
- Deleting old state without backup/recovery evidence.

DIAGNOSE IN THIS ORDER

1) Restate input/output grain or state boundary. 2) Compare expected vs actual. 3) Run the independent control. 4) Locate the first invariant that fails. 5) Repair only that boundary. 6) Re-run the original attempt for understanding, but preserve attempt_001 unchanged.

RECOVERY

Restore last known committed checkpoint, inspect partial side effects, make write idempotent/deduplicated, replay from safe boundary, validate, then commit state.

WHY THE WORKING PATTERN IS SOUND

The old checkpoint remains the source of resume truth until durable output is proven. Atomic replacement avoids half-written checkpoint files.

CLOSE REVIEW - CHANGED RETRY

Move the injected failure to a different pre-commit point and re-prove safety.

MASTER WHEN

Changed retry succeeds without this review open; independent control passes; you can teach the concept and one failure/recovery path without reading.



R14 - Review C3-01-L14

FIRST CHECK

pytest includes failure/edge cases; source accounting reconciles; keys/temporal bounds checked; live state hash unchanged; backfill state clearly separate.

COMMON FAILURE SIGNALS

- Only happy-path tests.
- Asserting output produced by the same buggy function without independent control.
- Backfill reusing live cursor file.
- Calling green exit code "data validated."

DIAGNOSE IN THIS ORDER

1) Restate input/output grain or state boundary. 2) Compare expected vs actual. 3) Run the independent control. 4) Locate the first invariant that fails. 5) Repair only that boundary. 6) Re-run the original attempt for understanding, but preserve attempt_001 unchanged.

RECOVERY

When test/control fails, freeze source + state, find first violated invariant, repair that boundary, rerun changed fixture; never "fix" expected values to match bad output.

WHY THE WORKING PATTERN IS SOUND

Tests guard known behavior; reconciliation checks data accounting independently; state isolation prevents a historical operation from moving the live resume boundary.

CLOSE REVIEW - CHANGED RETRY

Run a changed historical window with one deliberate invalid record and re-prove all invariants.

MASTER WHEN

Changed retry succeeds without this review open; independent control passes; you can teach the concept and one failure/recovery path without reading.